

SRI SATYA VENKATA VINAY MAHESH PATNALA

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PROFESSIONAL SUMMARY

Robotics and Electronics Engineer with engineering foundations built across two countries B. Tech in Electronics & Communication Engineering in India and an MSc in Robotics at the Edinburgh Centre for Robotics, Heriot-Watt University. Independently conceived and delivered a full industrial inspection pipeline using NeRF and 3D Gaussian Splatting as the sole researcher on the project. Experienced across the full robotics stack: computer vision, deep learning, autonomous manipulation, human-robot interaction, and hardware with hands-on project experience in Python, PyTorch, COLMAP, ROS, and Arduino. Member of the Edinburgh Centre for Robotics and Heriot-Watt Maker Society.

EDUCATION

MSc Robotics

Jan 2025 – May 2026

Heriot-Watt University, Edinburgh | Edinburgh Centre for Robotics

- **Dissertation:** Defect Detection in Complex Environments using NeRF & 3D Gaussian Splatting supervised by **Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi**.
- **Modules:** Robotic Mechanical Systems, Machine Vision, Robotic System Science, Intelligent Robotics, Human-Robot Interaction.

B. Tech Electronics & Communication Engineering

June 2018 – July 2022

Vel Tech Rangarajan Dr. Sagunthala R&D Institute, Chennai, India

- **Published:** “Design of Printed Antenna for Next Generation Wi-Fi Devices” -International Conference on Emerging Technologies, Chennai, May 2022.
- **Core focus:** Embedded systems, Arduino, AI, and hardware prototyping.

TECHNICAL SKILLS

3D Reconstruction & Vision: NeRF, 3D Gaussian Splatting, COLMAP, Nerfstudio 1.1.5, Point Cloud Processing, Chamfer Distance, ICP, OpenCV

Deep Learning & AI: PyTorch, CNN Architectures, Ollama LLM, CUDA GPU Programming, Google Colab

Robotics & Simulation: ROS, Webots, Robot Kinematics, HRI Design

Programming & Tools: Python, C, MATLAB, Bash/Linux, ffmpeg, Git, VS Code

Embedded & Hardware: Arduino, ESP32-C3, FreeRTOS, ESP-IDF, oscilloscope verification, 3D printing, laser cutting

KEY ACADEMIC PROJECTS

Defect Detection via NeRF & Gaussian Splatting - MSc Dissertation

Supervised by Dr. Maria Koskinopoulou & Dr. Abderrahim Halimi

- **Architected** a full inspection pipeline from scratch, raw video to ffmpeg frame extraction, COLMAP pose estimation, Nerfstudio model training, point cloud export, and automated defect quantification.
- **Evaluated** Nerfacto vs Splatfacto side-by-side across PSNR, SSIM, LPIPS, and FPS on identical HDD surface datasets; Splatfacto reached 38.28 dB PSNR, 0.98 SSIM, and 27 FPS, confirming it as the superior approach for real-time industrial inspection.
- **Engineered** dual defect detection strategies in parallel: a 2D pipeline using illumination-normalised difference maps and heatmaps for visual anomaly localisation, and a 3D pipeline applying weighted Chamfer Distance (0.6x geometric + 0.4x colour) between clean and defective point clouds.
- **Quantified** surface defects Developed parallel 2D and 3D defect detection strategies; the 2D pipeline employs illumination-normalised heatmaps and difference maps for surface anomaly visualisation, while the 3D pipeline applies Chamfer Distance-based point cloud comparison between clean and defective Splatfacto reconstructions to quantify geometric deviations.

4-DOF Autonomous Robotic Arm - Robotic Mechanical Systems

Supervised by Dr. Xianwen Kong

- **Designed** and 3D-printed a complete 4-DOF robotic arm from mechanical concept to physical assembly, then implemented full forward and inverse kinematics in MATLAB using DH convention and Jacobian analysis.
- **Integrated** an OpenCV-based vision system for object detection and localisation, enabling closed-loop pick-and-place operations with real-time Arduino actuation connecting perception, planning, and hardware execution in a single pipeline.

Cheer Mate: HRI Fitness Coach - Human-Robot Interaction.

Supervised by Dr. Christian Dondrup

- **Programmed** a Marty V2 robot as a context-aware fitness coach, integrating an Ollama LLM for natural language dialogue and designing multimodal feedback (speech + physical movement) to adapt coaching style to user responses.
- **Built** a QR-code activity tracking system enabling personalised session management across multiple users taking the system from single-interaction demo to a repeatable, user-aware coaching experience.

ESP32-C3 Embedded Waveform Generator - Embedded Software

Supervised by Dr. Kieran O'Leary

- **Developed** a dual-signal firmware application on ESP32-C3 using Arduino IDE and ESP-IDF (FreeRTOS), generating two configurable periodic signals (DATA & SYNC) with timing parameters uniquely derived from surname-based specifications.
- **Verified** all signal outputs on an oscilloscope, confirming timing accuracy (T_{ON} 1100–1600 μ s, 11 pulses, 6500 μ s idle) against the given specification.

EXPERIENCE

Python & Data Science Intern

Feb 2024 – April 2024

EI Systems Services, India

- Built foundational Python skills through applied data science tasks practised data manipulation, analysis, and visualisation using pandas and matplotlib on real datasets.
- Implemented a supervised machine learning model to predict insurance claims outcomes, gaining first hands-on experience applying ML concepts to a real business problem.

Customer Service Assistant

April 2025 – Present

Pasta Evangelists, Edinburgh, UK

- **Sustained** full-time MSc study alongside part-time shift work in a fast-paced production environment, demonstrating consistent workload management under competing demands.
- **Operated** within a diverse, multilingual team adhering to strict food safety and quality standards cross-cultural collaboration and regulated-environment discipline transferable directly to engineering team settings.

PUBLICATIONS & ACHIEVEMENTS

- Conference paper: “Design of Printed Antenna for Next Generation Wi-Fi Devices” - International Conference on Emerging Technologies, Chennai, May 2022.
- Affiliated researcher, Edinburgh Centre for Robotics (ECR) -one of the UK’s leading robotics research hubs.

EXTRACURRICULAR & LEADERSHIP

- **Contributed** as an active member of Heriot-Watt Maker Society used 3D printers, laser cutters, and resin printers to prototype physical components for robotics coursework, bridging classroom design and fabrication.
- **Participated** in inter-society hardware/software collaborative builds and 3D modelling workshops, taking initiative on projects outside assessed coursework.
- **Navigated** a full cross-country academic and professional transition from India to the UK adapting to a new research environment, institution, and engineering culture while delivering MSc-level research output.